

# Pencoedtre Fitness RPG – Game Design Document (Prototype v1)

## 1. Core Concept

Top-down 2D exploration RPG built with React + Phaser.

Students explore a lockdown version of Pencoedtre High School where an AI security system has trapped teachers and sealed departments.

To restore the school they must exercise, complete quests, defeat bosses, and unlock new zones.

Exercise is the primary gameplay mechanic.

## 2. Core Gameplay Loop

Explore School → Find NPC / trapped teacher → Receive Exercise Quest → Complete Exercise → Deal Damage / Gain XP → Defeat Boss → Unlock Department → Continue Story

## 3. Story Narrative

The player leaves class heading to the changing rooms or toilets. Suddenly lights turn red, alarms sound and metal shutters crash down as the lockdown system activates.

A tannoy announcement states that security lockdown has been initiated and everyone must remain in classrooms.

The player is trapped outside the classroom system and must restore power to the school by defeating bosses and rescuing teachers.

## 4. World Structure

The school is divided into biomes. Fog of war hides unexplored areas and defeating bosses unlocks the next zone.

Prototype zones:

Tutorial – Changing Rooms / Toilets

Zone 1 – PE Department (Sports Hall, Fitness Suite, MUGA)

Zone 2 – Science Block

Future zones may include Humanities, ICT, Art Department, Canteen and Library.

## 5. Character System

Players allocate starting attribute points:

Strength

Cardio

Power

Recovery

Example build:

Strength 5 / Cardio 2 / Power 2 / Recovery 1

## 6. Exercise to Combat System

Exercise translates directly into damage.

Example:

1 rep = 5 damage

1 kcal = 1 damage

Examples:

Squats = strength damage

Pushups = strength damage

Running/Bike/Skipping = cardio damage

Burpees = critical attack

Cool down = recovery

## 7. Boss Mechanics

Bosses have HP, weaknesses, resistances and hidden weaknesses.

Example Boss:

Sedentary Security Drone

HP: 200

Weakness: Cardio

Resistance: Strength

Hidden Weakness: Skipping

## 8. Preparation System

Students can prepare before boss battles by completing exercises which grant buffs such as attack boosts, stamina bonuses or dodge chance.

## 9. Recovery System

Cool down activities restore HP and stamina after workouts.

Examples: light cycling, walking, stretching.

## 10. Quests

Rescued teachers give quests which unlock further areas and workouts.

Example:

Mr Martin – Restore power to the Fitness Suite by completing 3 strength workouts.

## **11. Literacy and Numeracy Integration**

Quizzes and puzzles can unlock cosmetic rewards or buffs.

Examples:

Science – calculate watts produced

Maths – calculate damage from exercises

English – lore puzzles and story fragments

## **12. Reward System**

Rewards are UI-based rather than changing character sprites.

Examples include school badges, titles, pets, and trophies for the trophy cabinet.

## **13. Leaderboards**

Examples:

Most Workouts

Top Strength

Top Cardio

Boss Slayer

Weekly Champion

Future expansion: class leaderboards.

## **14. Player Controls**

Chromebook focused controls:

WASD movement

E interact

Space action

Touch controls can be added later for mobile.

## **15. Authentication**

Firestore authentication using school login.

Stores player stats, XP, quests and workout logs.

## **16. Art Style**

Top-down pixel RPG style similar to Stardew Valley or classic Pokémon.

Reasons:

Easy to build

Lightweight for browsers

Large asset libraries available.

## **17. Season Structure**

Each school half-term (6 weeks) acts as a game season with a final boss, leaderboard reset and new storyline.

## **18. Prototype Scope**

Movement

NPC interaction

One quest

One boss

Exercise logging

Zone unlock system

## **19. Development Milestones**

Phase 1 – Engine (movement, map, NPC interaction)

Phase 2 – Exercise system (logging and damage conversion)

Phase 3 – Boss fight system

Phase 4 – Progression system (XP, stats, leaderboards)

## **20. Long Term Vision**

Smartwatch integration

Heart rate tracking

AI exercise detection

Class competitions

Expanded school map